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Comparative Evaluation of Wheat Varieties for Resistance to Bird Cherry-Oat Aphid (*Rhopalosiphum padi* L.) under Field Conditions

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ABSTRACT

Wheat (*Triticum aestivum* L.) is an important grain and staple food crop, which plays an important role in the economy of Pakistan. The attack of insect pests has a significant impact on plant life and grain yields in wheat crop resulting in a considerable loss of yields. The most important approach to reducing these losses is through production of pest resistant/tolerant varieties. Among the insect pests the aphid *Rhopalosiphum padi* is a main destructive pests of wheat crop which cause a considerable damage to wheat production in Sindh Pakistan. In the present investigation, various varieties of wheat resistance to *R. padi* under field condition were studied. Results of current study showed significantly maximum population of *R. padi* (13.54 ± 2.26 /tiller) was recorded on Kiran, followed by TJ-83 11.35 ± 2.12 /tiller, Sattar 7.23 ± 1.64 /tiller, Mehran 5.08 ± 1.13 /tiller, TH-83 4.33 ± 0.98 /tiller, NIA Sunder 2.25 ± 0.52 /tiller, NIA Amber 2.72 ± 0.60 /tiller, Khirman 3.17 ± 0.67 /tiller, NIA Sarang 3.61 ± 0.77 /tiller, while minimum population was found on TD-1 (1.63 ± 0.40 /tiller) genotype respectively. The results indicated that highest yield of wheat genotype was recorded from TD-1 (6249.5 kg) followed by NIA Sunder (4562.1 kg), TH-23 (4343.1 kg), Mehran (4311.5 kg), Khirman (4193.2 kg), NIA Amber (3466.1 kg), NIA Sarang (3306.8 kg), Sattar (2826.2 kg), TJ-83 (2400.5 kg) whereas the Kiran genotypes found lowest (2293.1 kg) yield respectively.

Key words: *Rhopalosiphum padi*, wheat, screening, varieties

INTRODUCTION

Wheat (*Triticum aestivum* L.) belongs to family Poaceae, one of the main cereal crop and staple food in Pakistan as well as around the world. There is approximately 40% of world population consumes wheat (Ahmad and Shaikh, 2003; Grote et al., 2021; Shahzad, et al., 2021) because of its superior makeup, which includes more vitamins, minerals, fibers, proteins, fats and calcium (Irge, 2017). There are 22 million acres under wheat cultivation and it contributes 1.8% of nations GDP and 7.8% of the value added in agriculture (Sameen, et al., 2023; Saeed, and Hammad, 2022). The area under cultivation during 2021-22 was 8,976 thousand hectares (2.1%) less than the 9,168 thousand hectares in previous year compare to the 27.464 million tonnes produced last year the wheat production decreases to 26.394 million tonnes (3.9%) (Hassan, et al., 2023). The decline trained in the area and production of wheat due to the lack of irrigation water and dry conditions during sowing, a decrease in fertilizer intake, increase heat wave (Kajla et al., 2015) most importantly lack of insect pest management through

integrated pest management (IPM) approaches were the main causes of the loss in wheat yield (El-Gizawy, 2009; Khattak, et al., 2007; Grasswitz, 2019). Under the field conditions, number of insect pests infesting the wheat crop includes termites, cutworm, army worm, wheat beetles, aphids, thrips, bugs, leaf hoppers, and Hessian flies (Miller and Pike, 2002; El-Wakeil and Volkmar, 2013). The aphid poses a significant economic risk of infestation to wheat crop due to its ability to both directly damage the crop by sucking sap from leaves and indirectly promote disease transmission (Bukvayova, et al., 2006; Ali, et al., 2015). The species of aphids found on wheat crop such as *Sitobion miscanthi*, *Rhopalosiphum padi*, *Diuraphis noxia*, and *Schizaphis graminum*, and *Rhopalosiphum maidis* (Inayatullah et al., 1993; Helmi, and Rashwan, 2013; Yang, et al., 2019). Aphids are small in size, soft bodied insects that are less than 7 mm in size. They are the most devastating insect pests that feed on plant sap, acting as phloem feeder. They also inject the toxic saliva that may be phytotoxic in plant (Dedryver, 2010). The aphids infestations significantly reduce the agricultural

productivity and profitability of cereal crops by inhabiting chlorophyll concentration, root growth and shoot growth of plant (Riedell and Kieckhefer, 1995). These are usually found tender terminal shoots or under side of leaves, they live in colonies. Aphids secrete a large amount of honeydew, a sugary liquid that normally turns them black due to sooty mold appearance on leave (Kennedy, and Storer, 2000). The ability of the resistant type of cultivars to reduce the aphid population on cereal crops. The benefits of integrated pest management (IPM) program greatly from the adoption of resistant wheat cultivars, which are particularly effective in controlling cereal pests like aphids (Khan and Port, 2008). The quantities of various biochemicals in wheat cultivars that are antibiotic which resistant and protect them against insect pests like aphids (Leszczynsk, et al., 1989). Important information about whether a plant is suitable for pests or not can be obtained from its many insect resistance variants (Mottaghinia et al., 2011). The primary source of resistance to many insect pests in wheat cultivars is the presence of genetic variation (Tadesse, et al., 2022). Farmers in Pakistan have begun spraying pesticides on their wheat crop, and the spray area is expanding, since aphids are also emerging as a possible insect pest and a danger to wheat crop productivity (Wains et al., 2014; Anonymous, 2011). Thus, the primary goals of the current study are to screen wheat for resistant varieties against *R. padi* on wheat crops.

MATERIALS AND METHODS

Location: The current study was conducted in 2022–2023 at the Plant Protection Research Institute, Agriculture Research Sindh, Tandojam. Plant Protection Research Institute's coordinates are latitude 25.423° N, longitude 68.537° E, and elevation 79, respectively.

Experimental Design: The experiment with three replications of each treatment was set up using randomized complete block design (RCBD). The all-agronomic practice was applied uniformly on all sowing replicates. The replicated plot measured 100 by 100 feet. For wheat crops, the suggested seed rate is 50 kg/acre, and the study maintained a line spacing of 30 cm.

Wheat varieties: The ten wheat genotype TD-1, TJ-83, Sattar, TH-23, Mehran, Kiran, NIA Sarang, Khirman, NIA Sunder and NIA Amber were used in study.

Data Collection: The data of aphid population were taken on weekly basis from sowing up to harvesting of crops. The ten tillers from each treatment per replication were randomly selected and observed the population of aphid on various genotype of wheat.

Statistical Analysis: Analysis of variance (ANOVA) and mean value comparison utilizing least significant difference (LSD) at $P > 0.05\%$ significant difference were used to statistically assess the collected data. The Correlation were estimated to determine the impact of various weather parameters on the *R. padi* population on various wheat varieties.

RESULTS

In the present study the ten varieties of wheat were used against *R. padi* population under field condition of Tandojam during 2022-23. The calculated mean values

indicate the *R. padi* population's increasing trend from the first week of December to the third week of February as the plant's growth stages were proceed. The results of study confirmed significant impact ($F= 92.90$; $P<0.05$) of various varieties, the population of *R. padi* was appeared on all wheat varieties from the 1st week of observation up to 12 week, then gradually decreased to wards maturity of crops. The data showed (Table 1.) the initial *R. padi* population (2.33 ± 0.64 /tiller) was recorded on Kiran, followed by TJ-83 1.33 ± 0.36 /tiller, Sattar 1.03 ± 0.36 /tiller, Mehran 0.90 ± 0.35 /tiller, TH-83 0.86 ± 0.26 /tiller, NIA Sarang 0.73 ± 0.40 /tiller, Khirman 0.66 ± 0.40 /tiller, NIA Amber 0.40 ± 0.13 /tiller, NIA Sunder 0.36 ± 0.23 /tiller and TD-1 0.23 ± 0.21 /tiller varieties of wheat. Afterwards, the population of *R. padi* was rapidly increased on different wheat varieties, the highest weekly mean population was recorded on 4th week of observation, The highest weekly mean population of *R. padi* was recorded on Kiran (28.10 ± 1.91 /tiller) while the lowest weekly mean population was recorded on TD-1 (4.83 ± 1.05 /tiller) followed by NIA Sunder (6.06 ± 1.43 /tiller), NIA Amber (6.90 ± 1.69 /tiller), Khirman (7.03 ± 1.38 /tiller), NIA Sarang (8.36 ± 1.93 /tiller), TH-23 (10.23 ± 1.68 /tiller), Mehran (12.46 ± 2.74 /tiller), Sattar (18.06 ± 3.10 /tiller) and TJ-83 (26.10 ± 3.45 /tiller) respectively. The data indicated that (Figure 1.) overall highest mean population of *R. padi* (13.54 ± 2.26 /tiller) was recorded on Kiran, followed by TJ-83 11.35 ± 2.12 /tiller, Sattar 7.23 ± 1.64 /tiller, Mehran 5.08 ± 1.13 /tiller and TH-83 4.33 ± 0.98 /tiller respectively. Whereas the lowest population (1.63 ± 0.40 /tiller) was found on TD-1 followed by NIA Sarang 3.61 ± 0.77 /tiller, Khirman 3.17 ± 0.67 /tiller, NIA Amber 2.72 ± 0.60 /tiller, NIA Sunder 2.25 ± 0.52 /tiller respectively. The significant impact of weather (Table. 2) all parameters of the weather are maximum and minimum temperature, relative humidity and wind speed showed their impact of population of *R. padi* on various genotype of wheat during present study. According to results (Table. 3), the maximum and minimum temperatures showed a positive correlation, the maximum ($r= 0.4201594$) and minimum ($r= 0.379403971$), while humidity and wind velocity showed negative correlate with population of *R. padi*, humidity ($r= -0.386887492$) and wind ($r= -0.264254285$) respectively. The appearance of *R. padi* population on different genotype also contribute significant ($F= 10.76$; $P<0.05$) towards the yield of different wheat genotype. The results of study indicated (Table 3.) that maximum yield of wheat was recorded in TD-1 (6249.5 kg) followed by NIA Sunder (4562.1 kg), TH-23 (4343.1 kg), and Mehran (4311.5 kg). Move over the minimum yield was recorded in Kiran (2293.1 kg) followed by TJ-83 (2400.5 kg), Sattar (2826.2 kg), NIA Sarang (3306.8 kg), Khirman (4193.2 kg) and NIA Amber (3466.1 kg) respectively

Table 1. Weekly mean population of *Rhopalosiphum padi* on different wheat varieties

Date	TD-1	TJ-83	Sattar	TH-23	Mehran	Kiran	NIA Sarang	Khirman	NIA Sunder	NIA Amber
05.12.2022	0.23±0.20o-p	1.33±0.36e-p	1.03±0.36f-p	0.86±0.26i-p	0.90±0.35h-[	2.33±0.64b-p	0.73±0.40k-p	0.66±0.40k-p	0.36±0.23n-p	0.40±0.13n-p
12.12.2022	0.53±0.34l-p	2.03±0.83c-p	2.000±0.68c-p	1.03±0.40f-p	1.16±0.50e-p	3.83±0.98w-p	0.96±0.44g-p	0.86±0.38i-p	0.63±0.23k-p	0.76±0.32j-p
19.12.2022	1.96±0.89c-p	10.30±1.84i-p	7.43±1.24n-x	5.00±1.75t-f	5.66±1.47r-c	13.80±1.72f-j	4.33±1.89v-n	3.63±1.53w-p	2.73±0.83z-p	3.00±0.97y-p
26.12.2022	4.83±1.05t-i	26.10±3.45ab	18.06±3.10de	10.23±1.68i-p	12.46±2.74h-l	28.10±1.91a	8.36±1.93m-u	7.03±1.38n-x	6.06±1.43q-b	6.90±1.69n-y
02.1.2023	3.46±0.98x-p	22.16±4.07bc	16.73±3.15efg	9.96±1.89j-q	11.10±1.69i-m	24.20±2.88abc	7.43±1.74n-x	6.66±1.24p-z	4.96±1.02t-g	5.13±0.76s-e
09.1.2023	3.0±1.16y-p	17.13±3.11d-f	13.10±2.73g-k	7.63±1.87n-w	9.13±1.63k-s	21.06±3.29cd	6.16±1.71q-b	5.50±1.20s-d	3.86±0.85w-p	4.13±0.84w-p
16.1.2023	1.93±0.81c-p	14.06±2.77e-i	9.53±2.05k-r	6.43±1.59p-a	7.13±1.42n-x	17.60±3.05def	5.76±1.68r-c	4.83±1.22t-i	2.76±0.67z-p	4.46±0.82t-l
23.1.2023	1.26±0.62e-p	12.60±1.62h-k	6.63±1.33p-z	4.43±0.77u-m	4.90±0.72t-h	15.16±2.89e-h	4.23±0.59w-o	4.06±1.10w-p	2.00±0.57c-p	3.86±0.63w-p
30.1.2023	0.90±0.66h-p	10.90±2.38i-n	4.76±0.97t-j	2.90±0.54y-p	3.96±0.79w-p	12.76±2.75g-k	2.40±0.91b-p	2.26±0.90b-p	1.93±0.43c-p	2.03±0.62c-p
06.2.2023	0.73±0.38k-p	8.46±1.86l-t	3.96±0.77w-p	1.96±0.53c-p	2.46±0.56a-p	10.73±1.81i-o	1.63±0.76d-p	1.46±0.75e-p	1.10±0.38f-p	1.13±0.36e-p
13.2.2023	0.43±0.34m-p	6.86±1.73o-y	2.40±0.63b-p	1.03±0.45f-p	1.23±0.46e-p	8.26±1.75n-v	0.96±0.38g-p	0.73±0.50k-p	0.43±0.28m-p	0.56±0.35l-p
20.2.2023	0.26±0.17n-p	3.93±1.14w-p	1.16±0.56e-p	0.53±0.16l-p	0.83±0.27i-p	4.63±1.75t-k	0.40±0.14n-p	0.36±0.13n-p	0.20±0.16p	0.26±0.11op

*Mean followed by same letter in same Column is not significant difference (LSD,P<0.05)

Table 2. Correlation of weather parameters on population of *Rhopalosiphum padi* wheat varieties

Maximum temperature	r= 0.420159442; P>0.05
Minimum temperature	r= 0.379403971; P>0.05
Wind	r= -0.264254285; P>0.05
Humidity	r= -0.386887492; P>0.05

Table 3. Yield /acre of different wheat varieties

Varieties	Yield kg/h ⁻¹	Mean Group
TD-1	6249.5	a
TJ-83	2400.5	fg
Sattar	2826.2	efg
TH-23	4343.1	bc
Mehran	4311.5	bcd
Kiran	2293.1	g
NIA Sarang	3306.8	def
Khirman	4193.2	bcd
NIA Sunder	4562.1	b
NIA Amber	3466.1	cde

*Mean followed by same letter in same Column is not significant difference (LSD, P<0.05)

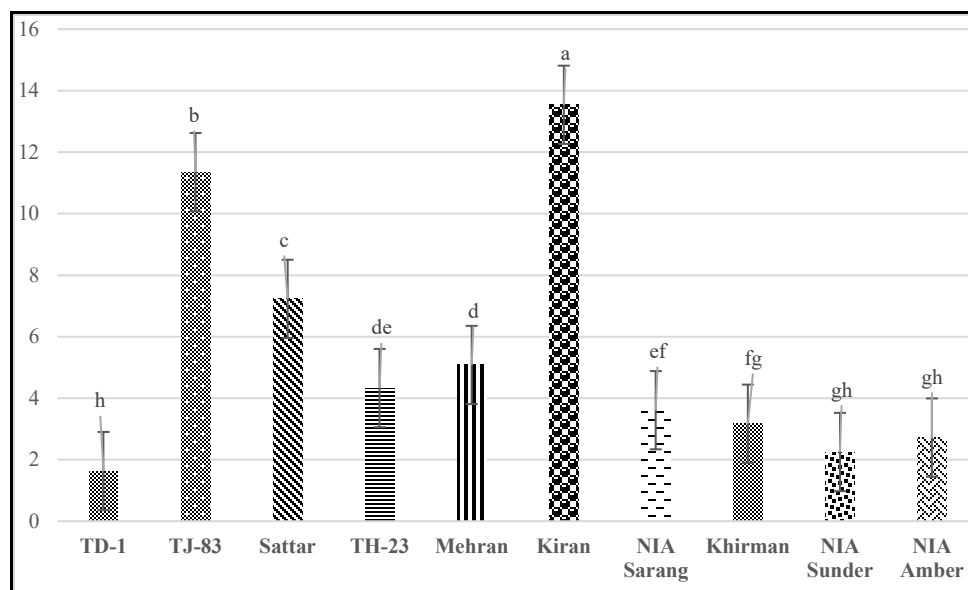


Figure 1. Overall mean population of *Rhopalosiphum padi* on different varieties of wheat

DISCUSSION

The aphid is becoming a more significant pest of wheat crops and could endanger Pakistan's wheat harvest. They directly lower grain quality and yield by sucking the plant's cell sap (Kirkland, et al., 2018). Wains, et al., (2014) mentioned the significant crop losses in many crops, including wheat, are caused by aphid infestation. In this regards the present study focused on the screening of the ten wheat genotype against aphid *R. padi* on under field condition. The population showed significant variation among the varieties to suffer from the attack of *R. padi*. Although the non of variety was found completely resistance but level of infestation differ considerably as TD-1 was less attack by *R. padi* followed by NIA Sunder, NIA Amber, Khirman, NIA Sarang, TH-23, Mehran, Sattar and TJ-83 varieties. The variety Kiran was found highly susceptible among the varieties against *R. padi*, as maximum population was recorded on it. Jan, et al. (2018) studied to evaluate the different wheat varieties for their resistance against aphid, however the Punjab 2011 showed comparatively resistant and variety Shafaq 2006 showed highly susceptible. In the current study it was observed that the population *R. padi* appeared on all genotype of wheat after the third week of sowing and gradually increased according to vegetative stages of crops. How ever some variation occurred in *R. padi* population, which might be due to resistance response of wheat genotype. The initial population of *R. padi* was appeared during the the first week of December and reached peak during fourth week of observation. The study also highlighted lack of resistance varieties of

wheat against aphid and emphasis on breeder and researcher develop such varieties to boost the production of wheat (Wains, et al., 2014). Rustamani, et al., (1999) reported that the initial infestation of aphid recorded during the third week of December and maximum population was found up to the month of March. Similarly, Ramzan, et al., (2020) revealed that the population of *R. padi* was reached peak from mid-January to mid-April on wheat genotype. Results showed that the initial highest population of *R. padi* (2.33 ± 0.64 /tiller) was observed on Kiran and lowest population was recorded on TD-1 (0.23 ± 0.20 /tiller) in the 1st week of December. According to Saleem, et al., (2009) The last week of December marked the beginning of the early aphid infestation, which peaked in the first week of March after remaining low during January. Similarly the Ali et al., (2015) stated that while the vegetative development continued, the aphid infestation grew, beginning in the first week of February, while the highest weekly mean population of *R. padi* (28.10 ± 1.91 /tiller) was observed on Kiran and lowest population (4.83 ± 1.05 /tiller) was found on TD-1 followed by followed by NIA Sunder (6.06 ± 1.43 /tiller), NIA Amber (6.90 ± 1.69 /tiller), Khirman (7.03 ± 1.38 /tiller), NIA Sarang (8.36 ± 1.93 /tiller), TH-23 (10.23 ± 1.68 /tiller), Mehran (12.46 ± 2.74 /tiller), Sattar (18.06 ± 3.10 /tiller) and TJ-83 (26.10 ± 3.45 /tiller) respectively. The findings of Zeb, et al., (2011) confirmed that the wheat line reported the largest mean population of aphids overall per plant on genotype PR-1 (42.20); these were followed by Khber-87 (32.57), Pirsabak -2005 (29.36) and KT-2004 (21.22 aphids per

plant), Saleem-2000 (19.92 aphids per plant), Pirsabak-2008 (15.71 aphids per plant), PR-2 (13.34 aphids per plant) and Pirsabak- 2004 (12.89 aphids per plant). Al-Hussine, and Alyousuf, (2021) additionally revealed that the varieties IPA-95 and Abu-Ghraib had the highest population rates of *R. padi*, averaging 139.83 and 169.31 aphids plant⁻¹, respectively, while the varieties Bhooth-22, Bhooth-158, and IPA-99 had the lowest rates of population density (89.03, 99.41, and 100.25 aphids plant⁻¹, respectively) in comparison to the other varieties. Aslam et al., (2005) examined twelve wheat varieties/lines against aphids and reported that all varieties/lines were attacked by aphids. Aphid attacks were lowest on the Inqalab-91 variety of wheat and greatest on the PND-1 variety/line. The current study's findings indicated that, in contrast to humidity and wind velocity, which exhibited negative correlations with *R. padi* populations, maximum and lowest temperatures showed positive correlations with it respectively. Aheer et al., (2008) revealed that when the temperature hit both the maximum and minimum limits, the aphid population dropped. When the temperature was in an aphid-friendly range in March, the population increase of *R. padi* and *S. avenae* was considered to have reached its maximum. Additionally, we observed variations in the mean population of both aphids at various locations. These variations may have resulted from abiotic factors including humidity, temperature, and rainfall. Many variables influenced the sharp rise and fall in the aphid population (Aheer et al., 2008). Additionally, it has been discovered that a variety of physical and biological elements were responsible for changes in the densities of the aphid population (Naeem, et al., 1996). According to Aheer et al., (2008) and Aslam, et al., (2004) temperature have a crucial role in the fluctuation of the aphid population. Similarly, the abundance of aphid population was also affected owing to the heavy rainfall (Naeem, et al., 1996). The results showed that *R. padi* population significantly impact on the yield of wheat varieties. The results indicated that maximum population of *R. padi* (13.54±2.26/tiller) was recorded on Kiran which showed the lowest yield 2293.1 kg, while the minimum population (1.63±0.40) was observed on TD-1 which showed highest yield 6249.5 kg/acre respectively. Shahzad et al., (2013) revealed that there were very substantial differences in the aphid population for both wheat yield and thousand grain weight. Shahbaz, et al., (2020) showed that early sowing dates produce a drop in production because aphids feed on grains, reducing their weight. This is because the climate encourages the growth of aphids.

CONCLUSION

Based on finding of current study it is concluded that TD-1 showed resistant varieties of wheat against *R. padi*, while the Kiran varieties found susceptible against *R. padi*. Monitoring the *R. padi* population regularly on wheat varieties and *R. padi* resistant varieties should be grown for reducing the losses.

CONFLICT OF INTEREST

The author declares that there is no conflict of interest with the article published in this journal.

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